

Introduction

How does the brain work? What do we want computers to do? Do we want our computers and computer-enhanced systems to act the way we do? If so, how do we want them to mimic the abilities of humans?

Since World War II, a group of scientists have attempted to understand the human nervous systems and to build artificial systems that act the way we do, at least a little bit.

In this book, we have put together a series of interviews of well-known, productive scientific leaders in the important, controversial, mildly glamorous and very high-risk area of science described as "brain theory," "neural networks," or "theoretical neuroscience." The interviews describe how this science is done and how it was done in the words of the scientists doing it.

Many of the early developments of brain theory sprang from the roots of cybernetics. Other theorists took inspiration from neuroscience, physics, electrical engineering, mathematics, and even economics. The neural network field is a hotbed of cross-disciplinary activities, where these messengers from many disciplines and their ideas collide like subatomic particles swirling around a particle accelerator. The interactions, influences, and changes occasioned by these collisions are described in this collection of informal conversations.

One of us (Anderson) is a well-known scientist in the neural network field. The other (Rosenfeld) is a journalist who has chronicled neural net developments for more than a decade. Many of our interview subjects directed specific comments to one or both of us in our roles as a scientist who participated in some of this history and as a journalist who has described some of this history in print, particularly the parts related to practical applications, computer applications, and business developments.

What emerges are fascinating life stories: seventeen people at the center of complex scientific and social developments. Their personal stories include intimate and often touching details about their childhoods and their families that complement the details of their discoveries.

This book contains the words of individuals who agree about some things, but disagree about more. They range in age from the mid-thirties to the mid-seventies and were educated in several continents, although most of them

currently live in the North America. Most are academics. However, anyone who believes that academia is an island of serenity populated with tweedy Mr. Chipses instructing the young in the ways of man and the world with wise and prudent counsel has obviously had little to do with a modern university.

Bringing a new field into existence involves the participants in a bitter and sometimes brutal Darwinian struggle for jobs, resources, and reputation. All of our interviewees had been through this process. The first thing they discovered about trying to promulgate genuinely new ideas is that most of their colleagues were not really very interested in hearing about them. The participants' struggles simply to obtain access to scientific communications channels, so that others would hear what they had done, occupy a significant part of several interviews and form an undercurrent in others.

Our interviews present neural network stories—the stories that have been told, referred to, whispered about, and imagined throughout the history of the neurocomputing field. These interviews are a Rashomon-like web of reality slices, rendered by interested parties, raconteurs, and sometimes those who believe that they are, in fact, the central characters. What's more, some of the mythic people responsible for the foundations of modern brain theory, communications, and computing—such as Norbert Wiener, Warren McCulloch, and Frank Rosenblatt—appear prominently in some of the recollections collected here.

One way to view this collection of stories is as a set of candid discussions about how science is actually done, as opposed to how we are told it should be done. Sometime during high school, we get exposed to a bloodless something called the "Scientific Method," which consists of a set of rules for doing science the proper way. First, we generate a hypothesis. Second, we test the hypothesis by doing an experiment. Third, we modify or reject our hypothesis based on the results. Like most things we learn in high school, such an idea has only a vague connection to reality. This book is about the reality.

These interviews were done in hotel rooms, in offices, and, in one case, by telephone over a period of several years. We were interested in the history of the neural network field and of the ideas that formed it as well as in the personal history of the participants. We had a list of topics we wanted to cover, but we often did not need to ask our questions. All of our interviewees have extensive experience teaching and lecturing, and most of them shaped their own narrative, with only modest prodding from us. Looking over the transcripts, almost all our predesignated topics were covered in some form or another during each interview. After a couple of questions to get things going, most interviewees became self-propelled and developed their story as they wished.

Over and over again, we were struck by the drama and passion involved in the stories. The questions these scientists wanted to answer in their work were important because they involved the way the mind works, the way the

brain works—the way we as humans connect to and contact the physical world. Surprisingly, they sometimes involved religion, as something both to reject and to accept. None of our subjects fit the stereotype of scientists as cold-blooded, remote intellects. Exactly the opposite was the case. The work they were doing was an obsession and so dominated their lives that often they could think of little else.

Some of our interviewees worked alone, some with a few local colleagues. However, there were two particularly productive centers of work on neural networks and brain modeling. One was in San Diego in the early 1990s, a time and place that had a major impact on nearly half of our interviewees. The famous two-volume "PDP books," published in 1986, summarized the work of the "parallel distributed processing group" at the University of California at San Diego in this era.

However, by far the most romantic center of research in brain theory was the group that formed in the 1940s and 1950s around the charismatic Warren McCulloch and his brilliant, unstable, and ultimately tragic collaborator, Walter Pitts. Jerry Lettvin, Michael Arbib, and Jack Cowan describe this time in some detail and give their sometimes differing perspectives on it. The interviews are arranged in order of the scientist's date of birth so some of these historical themes can become clearer.

We would not want to give the impression that our interviews reflect only dour intensity. There are stories with considerable humor: Bernie Widrow sliding down the hills of San Francisco in his new, leather-soled shoes; Jerry Lettvin's future in medical school being determined by an unusual gross anatomy exam; Michael Arbib's misadventures job hunting in California.

This book will not provide an introduction to neural networks or brain theory, though most of the important ideas are quite accessible. For a formal introduction to the field, let us recommend two collections of well-known papers that we edited: *Neurocomputing* (1988) and *Neurocomputing 2* (1990). Anderson's textbook, *An Introduction to Neural Networks*, gives a systematic introduction. At this time, a number of other good introductions at many levels of complexity are in print.

Some of the interviews contain quite technical material. We have left much of this material in place, although there is no actual mathematics in this book. Mathematics is a powerful tool for working with complex systems, but, unfortunately, it often serves as a barrier rather than an aid to understanding. One way to view mathematics is as a kind of language that very accurately describes some important kinds of complexity. It is a difficult language to learn to speak. If a reader does not happen to speak that particular language, however, many of the basic ideas can still be communicated, often in simple sentences and images. Mathematics makes these intuitions precise and usable, but the essence of the ideas often comes through without the mathematics. In this context, some of the seemingly throwaway comments in the interviews are very deep. Some of these comments may strike resonances in

those knowledgeable in the field, but the basic issues should be quite clear to everyone. Real science is often like this: it starts with simple questions; insights often begin as fuzzy pictures or vague verbal descriptions; and deep intuitions are communicated offhandedly, using metaphor and analogy.

Interviews that touch on such deep issues include Carver Mead's discussion of analog versus digital processes and Jerry Lettvin's comments on the role of discrete and continuous processes in the thinking of Walter Pitts and on the way Leibniz influenced both Pitts's thinking and indirectly the entire McCulloch group. Many interviews touch on the important distinction between engineering ("it works") and science ("it works like the brain"). Often, what seem like far-fetched connections turn out not to be: Jack Cowan's discussion of the forms that hallucinations take and the underlying functional cortical neuroanatomy is a striking example.

These talks also mention the controversies that surrounded the rise and fall of public and scientific interest in Rosenblatt's perceptron, as well as the influence of Marvin Minsky and Seymour Papert on the loss of interest in neural networks during the 1970s. Robert Hecht-Nielsen addresses this history during his interview and also describes some of the history of the momentous IEEE neural net meetings that took place in the late 1980s, a subject also addressed in detail in Bart Kosko's interview.

Another controversy raised in a number of the conversations is the scientific "credit assignment" problem related to the backpropagation algorithm, discovered by a number of different researchers seemingly independently. There is also a highly charged area related to the tremendous public attention brought to the field in the early 1980s by two of John Hopfield's publications in the *Proceedings of the National Academy of Sciences*.

We hope that some of the excitement and power of the ideas about brain function and intelligent systems discussed in this book come across to others as powerfully as they did to us during the actual interviews.

Let us make a few technical comments. These interviews were tape-recorded and expertly transcribed by Dr. Emily Pickett of the Brown University Department of Cognitive and Linguistic Sciences. We owe Dr. Pickett a great debt. No one speaks grammatical English when talking out loud, even Ph.D.s, even for the record, so we edited the raw transcripts in order to make them reasonably grammatical, to eliminate repetitions and hesitations, and to tighten them up when interviewees started to ramble. We also eliminated much of the redundancy that characterizes a long conversation. We also had to work within length limitations forced upon us by the economics of book publishing. The editorial assistance and publishing guidance provided by Betty Stanton was invaluable. In addition, our editor-shepherds, Jerry Weinstein, Wendy Drexler, and Katherine F. Arnoldi made sure that these interviews successfully completed the journey from manuscript to published book.

In editing the interviews we tried hard to retain the distinct personality of each speaker and let them convey their ideas as clearly as possible. The

speakers were offered the opportunity to make corrections, check spellings, and, rarely, to make additions.

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Note: John Hopfield is mentioned in many of the interviews in this collection. Though we were unable to meet with Professor Hopfield in person, we did conduct an interview with him in 1996 by telephone. However, due to technical problems that interview could not be transcribed. We asked Professor Hopfield to redo the interview by telephone early in 1997, as this book was going into its final stages of production. Unfortunately, he declined. We regret the loss.

